ADVERTISING SALES PREDICTION

Machine Learning Project Report

Prepared for:

Fundamentals of Machine Learning (Practical)
BCA AI – Semester III
Academic Year 2026–27

Submitted by:

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Project Type: Individual Project

SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS & SCIENCE

BCA AI – Semester III

2. Certificate

SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS & SCIENCE

Bachelor of Computer Application (AI)

CERTIFICATE

This is to certify that Mr. DHRUV LALJIBHAI BHARTHI (Roll No. 565) and Mr. SUHAN JITESHBHAI KAKLOTAR (Roll No. 571), students of Bachelor of Computer Application (AI), have satisfactorily completed the individual practical Machine Learning project titled "Advertising Sales Prediction" for Semester III during the academic year 2026-27.

_____	_____
Supervisor / Examiner	Principal
Date: _____	Date: _____

3. Problem Statement

Advertising is an important way for companies to promote their products and services. Companies spend money on different advertising channels such as TV, Radio, and Newspaper. However, it is not always easy to understand how advertising spending affects product sales.

The main problem addressed in this project is to predict sales based on advertising budgets. Machine Learning techniques are used to study the relationship between advertising channels and sales. This project uses Linear Regression and Ridge Regression to predict sales using historical advertising data.

4. Objectives

The main objectives of this project are:

- To understand the advertising sales dataset.
- To check missing values and duplicate records.
- To detect and handle outliers using the IQR method.
- To perform Exploratory Data Analysis (EDA).
- To visualize advertising data using different graphs.
- To generate an automated EDA report using Sweetviz.
- To build a Linear Regression model.
- To compare Linear Regression and Ridge Regression.
- To evaluate model performance using R^2, MSE, and RMSE.
- To use cross-validation for model evaluation.

5. Dataset Description

The dataset used in this project contains 200 records and 5 columns. It contains information about advertising budgets and sales.

Column	Description
ID	Unique record number
TV	TV advertising budget
Radio	Radio advertising budget
Newspaper	Newspaper advertising budget
Sales	Sales value

Input Features: TV, Radio, Newspaper
Target Variable: Sales
Total Records: 200
Total Columns: 5
Dataset File: advertising_sales.csv
Dataset Source: Insert the actual source or website link.

6. Data Preprocessing

6.1 Missing Values: The dataset was checked using `df.isnull().sum()`. No missing values were found in the original dataset.

6.2 Duplicate Records: The dataset was checked using `df.duplicated().sum()`. No duplicate records were found.

6.3 Data Cleaning: The `drop_duplicates()` function was used to remove duplicate records. The `fillna()` function was used to fill missing numerical values with median values if required.

6.4 Outlier Detection Using IQR: The IQR method was used to detect outliers in the Newspaper column.

IQR = Q3 – Q1
Lower Bound = Q1 – 1.5 × IQR
Upper Bound = Q3 + 1.5 × IQR

Calculated values: Q1 = 12.75; Q3 = 45.10; IQR = 32.35; Lower Bound = -35.775; Upper Bound = 93.625.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Pandas, Matplotlib, and Seaborn to understand the distribution and patterns in the advertising dataset.

7.1 Histogram

The histograms show the distribution of TV, Radio, Newspaper, and Sales values.

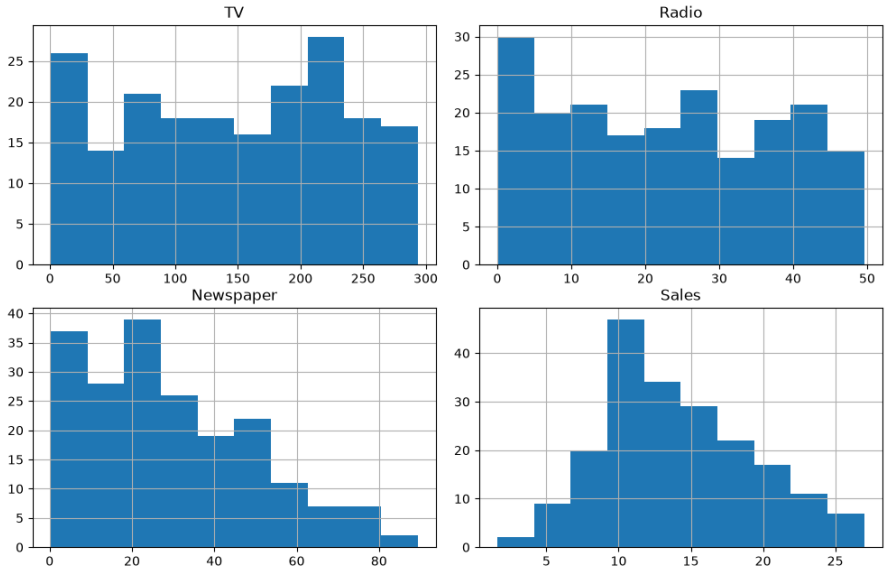


Figure 1: Histograms of TV, Radio, Newspaper, and Sales.

7.2 Box Plot

The box plot was used to identify outliers in the Newspaper column before and after outlier removal.

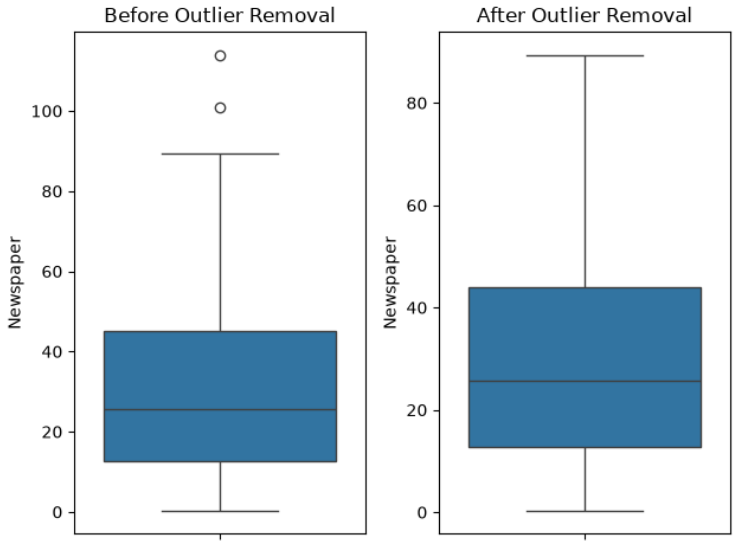


Figure 2: Newspaper Box Plot Before and After Outlier Removal.

7. Exploratory Data Analysis (continued)

7.3 Average Advertising Budget

The bar chart shows the average advertising budget for TV, Radio, and Newspaper.

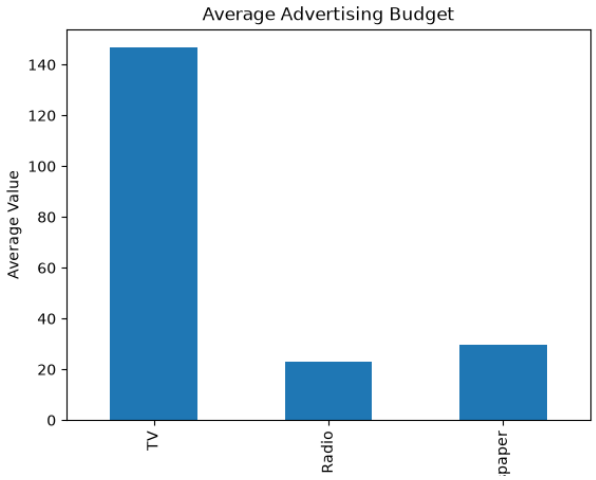


Figure 3: Average Advertising Budget.

7.4 Advertising Budget Distribution

The pie chart shows the percentage share of total advertising budgets across the three advertising channels.

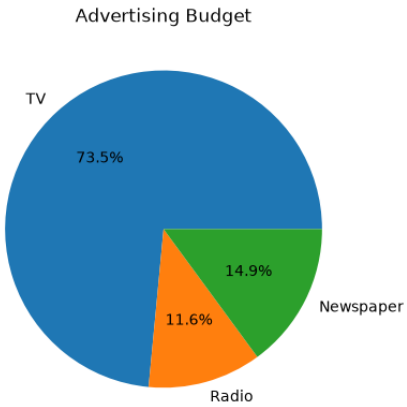


Figure 4: Advertising Budget Distribution.

7.5 Correlation Matrix

A correlation matrix was generated to understand relationships between numerical variables.

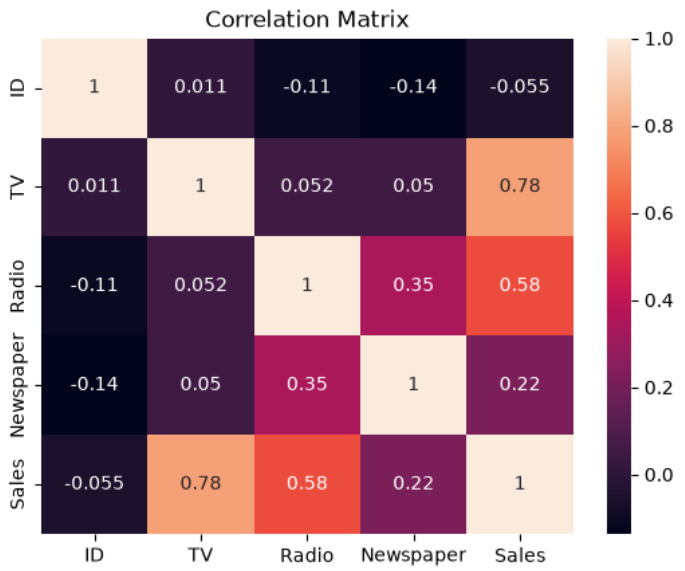


Figure 5: Correlation Matrix Heatmap. The displayed matrix shows a strong positive correlation between TV and Sales (0.78), and Radio and Sales (0.58).

7.6 Automated EDA Report

Sweetviz was used to generate an automated Exploratory Data Analysis report. The report includes dataset overview, column information, data distribution, statistical summaries, and feature relationships. The generated file is named `sweetviz_report.html`.

8. Feature Selection

Input Features (X): TV, Radio, Newspaper
Target Variable (y): Sales
The ID column was not selected because it is only a record identifier.

9. Model Building

The dataset was divided into training and testing data using an 80:20 ratio: 158 training records and 40 testing records.

Linear Regression: A supervised Machine Learning algorithm used to predict continuous numerical values. It was trained using TV, Radio, and Newspaper advertising budgets to predict Sales.

Ridge Regression: A regularized version of Linear Regression. It uses L2 regularization to reduce the effect of large model coefficients. Both models were trained and tested using the same train-test split.

10. Model Evaluation

The models were evaluated using three performance metrics.

R^2 Score: Measures how well the model explains variation in the target variable. A value closer to 1 generally indicates a better fit.

Mean Squared Error (MSE): Measures the average squared difference between actual and predicted values. Lower MSE indicates lower prediction error.

Root Mean Squared Error (RMSE): The square root of MSE. It measures prediction error in the same unit as the target variable.

11. Results

Linear Regression Results

Metric	Result
R^2 Score	0.898945
MSE	2.740912
RMSE	1.655570

Ridge Regression Results

Metric	Result
R^2 Score	0.898943
MSE	2.740985
RMSE	1.655592

Both models produced very similar results. Linear Regression performed very slightly better than Ridge Regression on the test dataset.

Actual vs Predicted Sales

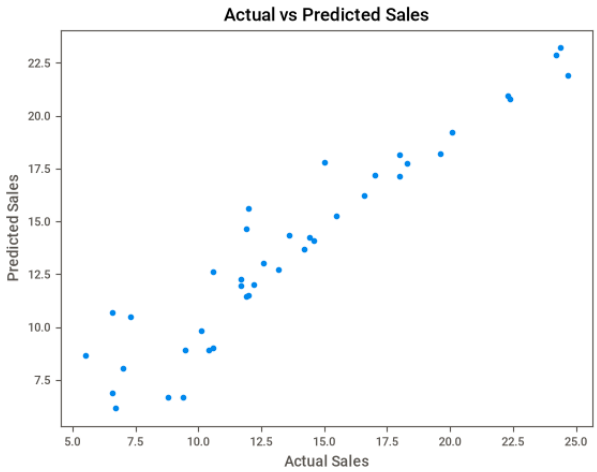


Figure 6: Actual vs Predicted Sales.

12. Cross-Validation

Five-fold cross-validation was used to check model performance across different data splits.

Cross-validation R^2 scores: 0.87988273, 0.91769650, 0.92510819, 0.80845813, and 0.89565733.

Average Cross-Validation R^2 Score: 0.885361

This gives an additional estimate of model performance across different data subsets.

13. Conclusion

This project successfully developed a Machine Learning model to predict sales using advertising budgets. The dataset was cleaned and analyzed using different data visualization techniques. Outliers in the Newspaper column were detected using the IQR method.

Linear Regression and Ridge Regression models were trained and evaluated. Linear Regression achieved an R^2 score of approximately 0.899, showing that the model explains a large part of the variation in sales in the test dataset. This project demonstrates how Machine Learning can help analyze advertising data and predict sales.

14. Future Scope

- Use a larger advertising dataset.
- Add more advertising-related features.
- Compare additional regression algorithms.
- Improve model performance through hyperparameter tuning.
- Develop a web application for sales prediction.
- Create a dashboard to display advertising data and predicted sales.

— End of Project Report —